

PROGRAM:

```
#1.a
import random
mylist=["Ann","Ben","ciril","John","Eric","Anu"]
random.shuffle(mylist)
print(mylist)
```

OUTPUT:

```
['Eric', 'John', 'Ann', 'Ben', 'ciril', 'Anu']
```

1.b

```
import time
print("Current time in sec :", time.time())
print("Current time:", time.ctime())
print("Current time after 30 sec:",
time.ctime(time.time()+30))
t=time.localtime()
print("local time :",t)
print("current day :",t.tm_mday)
print("current month :",t.tm_mon)
print("current year :",t.tm_year)
print("current hour :",t.tm_hour)
print("current minute :",t.tm_min)
print("current weekday :",t.tm_wday)
```

OUTPUT:

```
Current time in sec : 1735114367.16032
Current time: Wed Dec 25 08:12:47 2024
Current time after 30 sec: Wed Dec 25
08:13:17 2024
local time : time.struct_time(tm_year=2024,
tm_mon=12, tm_mday=25, tm_hour=8,
tm_min=12, tm_sec=47, tm_wday=2,
tm_yday=360, tm_isdst=0)
current day : 25
current month : 12
current year : 2024
current hour : 8
current minute : 12
current weekday : 2
```

#1.c

```
import calendar
month =int(input("Enter the month:"))
year =int(input("Enter the year:"))
print(calendar.month(year,month))
```

OUTPUT:

```
Enter the month:7
Enter the year:2025
    July 2025
Mo Tu We Th Fr Sa Su
   1  2  3  4  5  6
  7  8  9 10 11 12 13
 14 15 16 17 18 19 20
 21 22 23 24 25 26 27
 28 29 30 31
```

#1.d

```
import math
print("value of pi :",math.pi)
from math import sqrt ,pi
print("square root of 169:",sqrt(169))
print("value of pi :",math.pi)
print(math.cos(90))
print(math.sin(45))
```

OUTPUT:

```
value of pi : 3.141592653589793
square root of 169: 13.0
value of pi : 3.141592653589793
-0.4480736161291701
0.8509035245341184
```

#1.e

```
import statistics as s
mlist=[1,2,43,5,6,66,77,44]
print("Mean :",s.mean(mlist))
print("Mode :",s.mode(mlist))
print("Median :",s.median(mlist))
print("Harmonic mean
:",s.harmonic_mean(mlist))
print("Statistics variance:",s.variance(mlist))
print("Statistics Median low
:",s.median_low([40,55,74,7,-6,78,50]))
```

OUTPUT:

```
Mean : 30.5
Mode : 1
Median : 24.5
Harmonic mean : 4.122036430411067
Statistics variance: 956.2857142857143
Statistics Median low : 50
```

